

ity (though the opening of course lags behind the EMG activity). However, glottal abduction alone may not be enough to stop vocal-fold vibration (as we will see later), and there is some evidence that the CT may be used in voiceless stops (and fricatives) to increase longitudinal tension in the folds and thereby to prevent their vibration (see Hirose 1977; Löfqvist, McGarr, and Honda 1984; Löfqvist et al. 1989; also, Hutter 1985). This might explain why F_0 at the beginning of the immediately following vowel is usually higher for voiceless than for voiced stops (see e.g. House and Fairbanks 1953; Edwards 1981).

But we still have to consider the ensuing *adduction* of the vocal folds for the following vowel. After its peak, PCA activity basically just returns to the low level that it usually has in vowels. The activity of the IA, however, increases after its trough so vigorously as to form an extra peak before falling to the level normal for vowels (see, however, Löfqvist, McGarr, and Honda 1984: fig. 1). This peak of IA activity obviously has the function of quickly drawing the arytenoids (their posterior edges) back together so that modal vibration of the folds can begin. However, a corresponding albeit very low peak can sometimes be seen after *voiced* stops, even though the amount of preceding IA suppression in them is minimal (see Hirose and Gay 1972: fig. 3; Hirose and Ushijima 1978: fig. 1). This minor peak may serve to eliminate any pressure-induced separation of the arytenoids that might have occurred during the supralaryngeal closure. I will call such momentary peaks of extra adductor activity ‘hyperactivations’, whether there is a preceding suppression or not. (Note that for the sake of convenience, I will unorthodoxly regard not only the IA and the LCA but also the vocalis as an ‘adductor’ muscle.)

But contrary to Hirose and Gay’s (1972) conclusions, the LCA and the vocalis muscles might also play a role in the voicing distinctions. The data in Hirose 1977 (fig. 3) and Sawashima and Hirose 1983 (fig. 2-18) as well as those in Löfqvist, McGarr, and Honda 1984 suggest that this is the case in Swedish. In the voiced stops, the LCA and the vocalis are scarcely suppressed at all relative to the preceding vowel and then show a moderate peak at the beginning of the following vowel; in the voiceless stops, both muscles exhibit a clear suppression followed by a peak which mostly seems to be higher than that after the voiced cases. Hence the function of these muscles here appears to mirror, to some extent, that of the IA. The data of Hirose and Gay (1972), on the other hand, are too limited, especially in the case of the LCA, to establish that these muscles do not similarly contribute to the English voicing distinctions.

Unfortunately, none of the other studies (among those cited here) of

§ 3.3

Intrinsic laryngeal actions